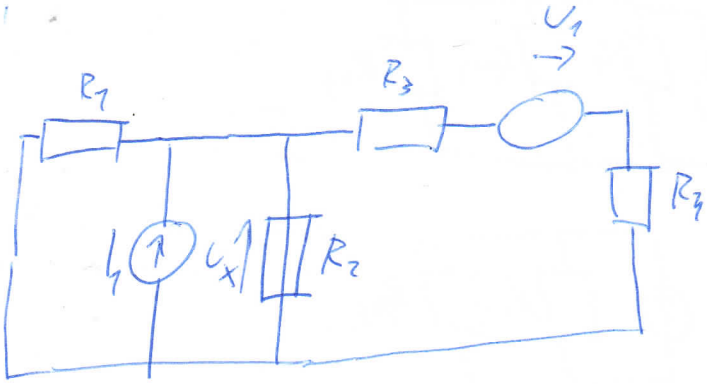
1

Tomáš Kysola



$U_1 = 12\,V$

$I_1 = 8\,mA$

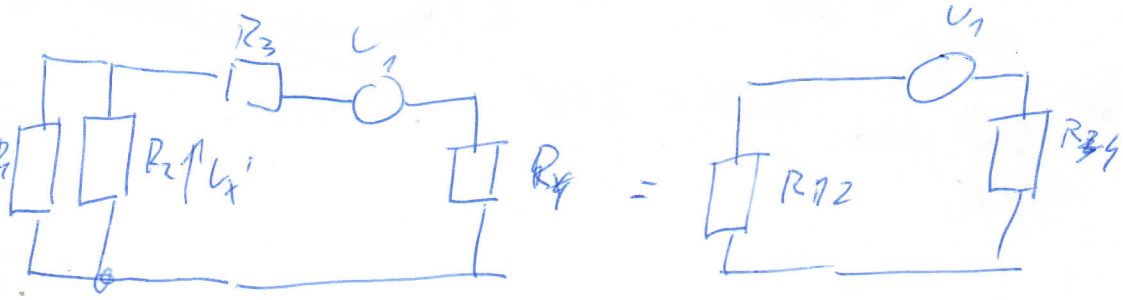
$R_1 = 15\,k\Omega$

$R_2 = 5\,k\Omega$

$R_3 = 250\,\Omega$

$R_4 = 1\,k\Omega$

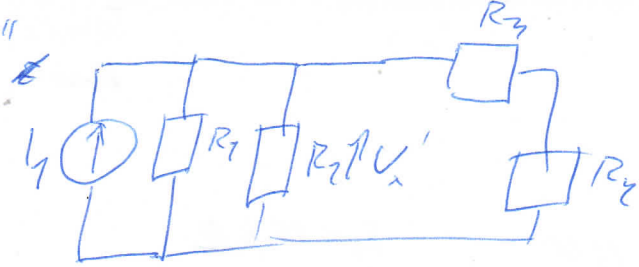
$\underline{U_x'}$ =



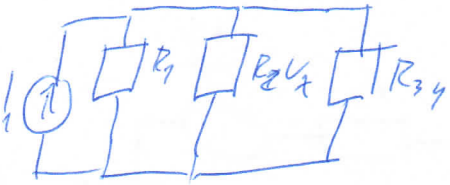
$$U_x' = U_1 \cdot \frac{R_{12}}{R_{12} + R_{34}} = U_1 \frac{\frac{R_1 \cdot R_2}{R_1 + R_2}}{\frac{R_1 \cdot R_2}{R_1 + R_2} + R_3 + R_4} =$$

$$= 12\,V \cdot \frac{\frac{15\,000 \cdot 5\,000}{20\,000}}{\frac{15\,000 \cdot 5000}{20000} + 250 + 1000} = 9\,V$$

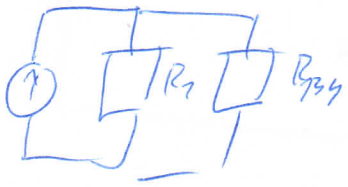
U_x''



$=$



$=$



$R_{34} = R_3 + R_4 = 1250\,\Omega$

$R_{134} = \frac{R_1 R_{34}}{R_1 + R_{34}} = \frac{15\,000 \cdot 1250}{15\,000 + 1250} = 1153\frac{11}{13}\,\Omega$

$\cancel{R_{34} = \frac{R_3 \cdot R_4}{R_3 + R_4} = \frac{250 \cdot 1000}{1250} = 200\,\Omega}$

$\cancel{R_{134} = \frac{R_1 \cdot R_{34}}{R_1 + R_{34}} = \frac{15\,000 \cdot 200}{15\,200} = 197\frac{7}{19}\,\Omega}$

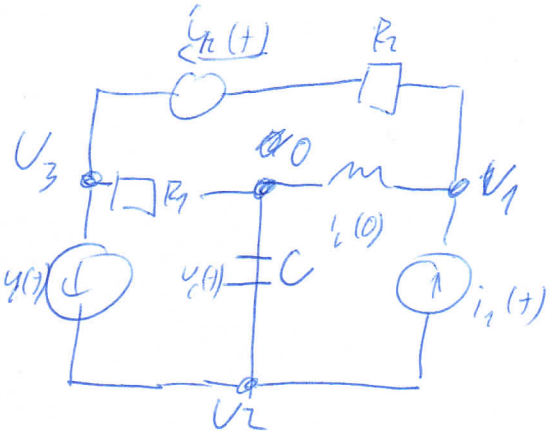
$U_x'' = I_1 \cdot \frac{R_{134} R_2}{R_{134} + R_2} = \frac{1153\frac{11}{13} \cdot 5000}{1153\frac{11}{13} + 5000} \cdot 0{,}008 = 7{,}5\,V$

$U_x = U_x' + U_x'' = 9V + 7{,}5V = \underline{\underline{16{,}5\,V}}$

(2)

$X_{MCN} = d - z_v = 4 - 1 - z_v = 4 - 1 - 2 = 1$

$X_{MSP} = s - z_i = 5 - 1 - 1 = 3 - 1 = 2$



$U_3 = u_2$

$z = u_1$

1:

$$\frac{1}{L}\int_0^t (u_1 - i_1)\,dt + i_1(0) - R_2 \cdot i_2(0) = 0$$